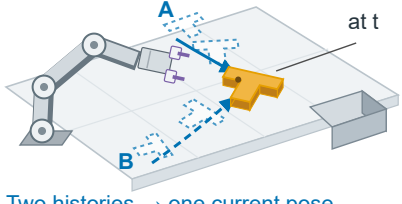
What must be remembered and predicted for interaction?

Conceptual illustration; all positions and trajectories are hypothetical

(a) Same pose, different motion



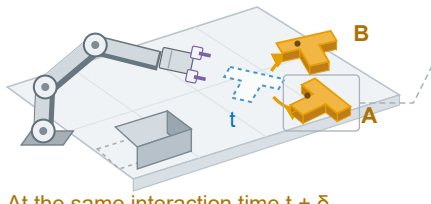
Two histories → one current pose

Past poses: t − 2Δ → t − Δ → t

—— A ---- B

Current pose alone is ambiguous.

(b) Different interaction poses



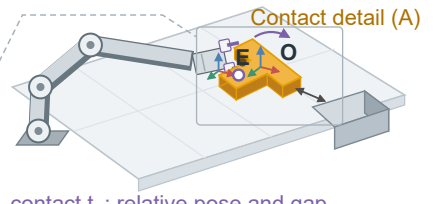
At the same interaction time t + δ

A / B → different position + rotation

Predict Δp, log ΔR and scene motion.

Carry the motion history forward.

(c) Resolve contact geometry



contact t_c: relative pose and gap

Target role: align the contact.

Context role: preserve clearance.

O: target; E: end-effector frame